

ENG-654 · LECTURE 02

URDF, Screw Motion & PoE

How can one robot model serve geometry, software, and the inverse-kinematics questions ahead?

First read the robot as a URDF tree; then recover its finite motion from screw axes and an ordered Product of Exponentials.

A readable robot needs complementary representations

The lecture follows one custom_3R robot from a compact table to a software model and a geometric motion formula.

1

DH

Which four parameters encode each deliberately assigned frame step?



2

URDF

How are topology, joint frames, geometry, and limits communicated to software?



3

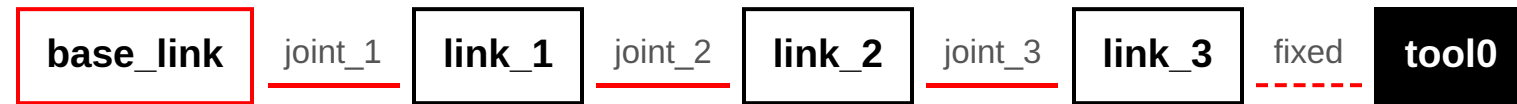
Screws and PoE

How does each joint generate motion, and how do those motions compose?

Invariant: after base, tool, and joint-zero conventions are aligned, every valid representation must predict the same forward kinematics.

A URDF robot is a rooted tree of links and joints

Read the parent–child structure first: it determines the unique path along which transforms are composed from the root to any tool or sensor.



Link

A named rigid-body frame with optional visual, collision, and inertial descriptions.

Joint

A directed edge from one parent link to one child link, with a type and motion rule.

Root

The only link with no parent joint; every other link must be reachable from it.

Joint origin and axis define one kinematic edge

At zero joint displacement, origin places the child link frame relative to the parent; motion then occurs about or along axis.

```
<joint name="joint_2" type="revolute">
  <parent link="link_1"/>
  <child link="link_2"/>
  <origin xyz="1 0 .5" rpy="0 0 0"/>
  <axis xyz="0 1 0"/>
  <limit lower="-3.14" upper="3.14"
          effort="100" velocity="2"/>
</joint>
```

Units and pose xyz uses metres; rpy uses radians. URDF roll–pitch–yaw gives $R_z(\psi)R_y(\theta)R_x(\phi)$.	Axis The direction is expressed in the joint frame. Normalise it before building a screw axis.
Types fixed, revolute, continuous, and prismatic are common; planar and floating support varies across consumers.	Range Limits constrain admissible joint positions, velocities, and efforts; continuous joints have no position bounds.

Kinematics positions link frames; property origins place their contents

Changing a mesh's visual xyz rpy can repair its appearance without changing any joint axis or the forward-kinematic chain.

<visual>

Appearance

A link may contain several visuals. Each geometry has its own origin, scale, colour, or material.

```
<visual>
  <origin xyz="0 .25 0" rpy="0 0
-1.5708"/>
  <geometry><mesh filename="link.stl"/>
</geometry>
</visual>
```

<collision>

Contact geometry

A link may contain several collision shapes, often simpler than the visual mesh.

```
<collision>
  <geometry><box size="1 .2 .2"/>
</geometry>
</collision>
```

<inertial>

Dynamics

A link has at most one inertial block: mass, centre of mass, and inertia tensor in its inertial frame.

```
<inertial>
  <mass value="2.4"/>
  <inertia ixx=".08" ixy="0" ixz="0"
            iyy=".11" iyz="0" izz=".09"/>
</inertial>
```

Mesh caveat: STL coordinates are unitless. Confirm the CAD export convention and use URDF scale when the mesh is not already expressed in metres.

A transmission describes an actuator interface, not kinematics

The root-to-tool transform is already determined by joint topology, types, origins, axes, and coordinates; transmission tags are interpreted by control or simulation software.

Illustrative XML excerpt; controller details depend on the software stack.

```
<transmission name="joint_2_drive">
  <type>transmission_interface/SimpleTransmission</type>
  <joint name="joint_2">...</joint>
  <actuator name="motor_2">
    <mechanicalReduction>80</mechanicalReduction>
  </actuator>
</transmission>
```

- **Kinematics:** links, joints, origins, axes, and joint coordinates.
- **Geometry:** separate visual and collision descriptions.
- **Dynamics:** mass, inertia, friction, and damping.
- **Actuation:** transmissions and software-specific control extensions.

Inspect the frame tree, then edit one joint origin

This laboratory reads link and joint topology plus origin transforms at $q = 0$. It intentionally does not render axes, joint motion, limits, or visual/collision/inertial content.

Explore: add a link and parent joint, edit an xyz value, or drag a non-root frame. Observe how the source and home-frame rendering remain synchronised.

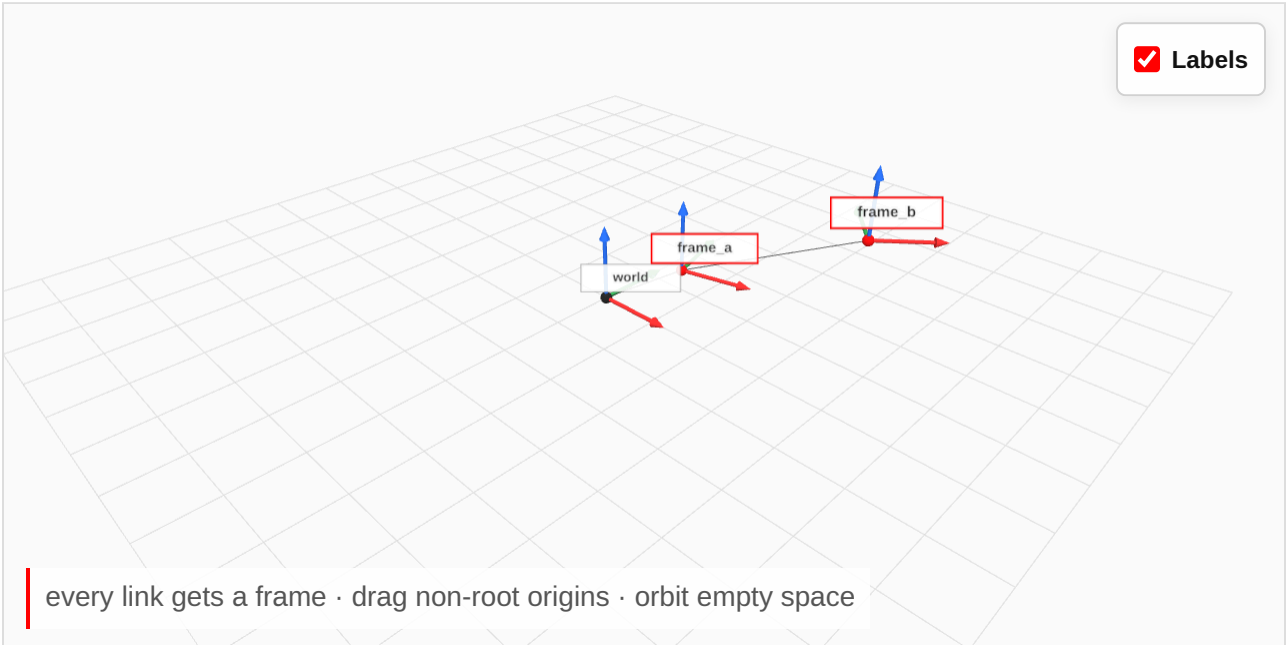
Editable URDF

Upload URDF

```
<robot name="two_frame_lab">
  <link name="world"/>
  <link name="frame_a"/>
  <link name="frame_b"/>

  <joint name="world_to_a" type="fixed">
    <parent link="world"/>
    <child link="frame_a"/>
    <origin xyz="0.60 0.20 0.40" rpy="0 0 0.25"/>
  </joint>

  <joint name="a_to_b" type="fixed">
    <parent link="frame_a"/>
    <child link="frame_b"/>
    <origin xyz="1.20 0.45 0.35" rpy="0 0 0.45"/>
  </joint>
</robot>
```



every link gets a frame · drag non-root origins · orbit empty space

frame_a.z  0.40

Fix the twist convention before calculating motion

This lecture uses angular-first coordinates $\xi = [\omega; \mathbf{v}]$; some texts and software reverse that order.

Frames and configuration

W : fixed world frame.

B : moving body frame.

q_i : displacement from the declared joint zero.

θ : scalar exponential coordinate—radians for a normalised rotational screw, metres for a normalised prismatic screw.

Twist quantities

$\omega \in \mathbb{R}^3$: angular twist component; a unit axis for a revolute or helical screw and zero for a prismatic screw.

$\mathbf{v} \in \mathbb{R}^3$: linear component of a rigid-body twist.

$\xi = (\omega, \mathbf{v}) \in \mathbb{R}^6$: twist coordinates.

Operators

$[\omega]_{\times}$: skew matrix satisfying $[\omega]_{\times} \mathbf{p} = \omega \times \mathbf{p}$.

$\hat{\xi} \in \mathfrak{se}(3)$: matrix form of a twist.

$\exp(\cdot)$: matrix exponential.

Finite pose

$R \in SO(3)$: orientation.

$\mathbf{t} \in \mathbb{R}^3$: translation.

$T \in SE(3)$: homogeneous transformation.

M : home pose at the chosen joint zeros.

Constant angular velocity naturally produces an exponential

Before using a 3×3 or 4×4 matrix exponential, follow one point and one frame rotating in the plane.

Define the 90° rotation generator

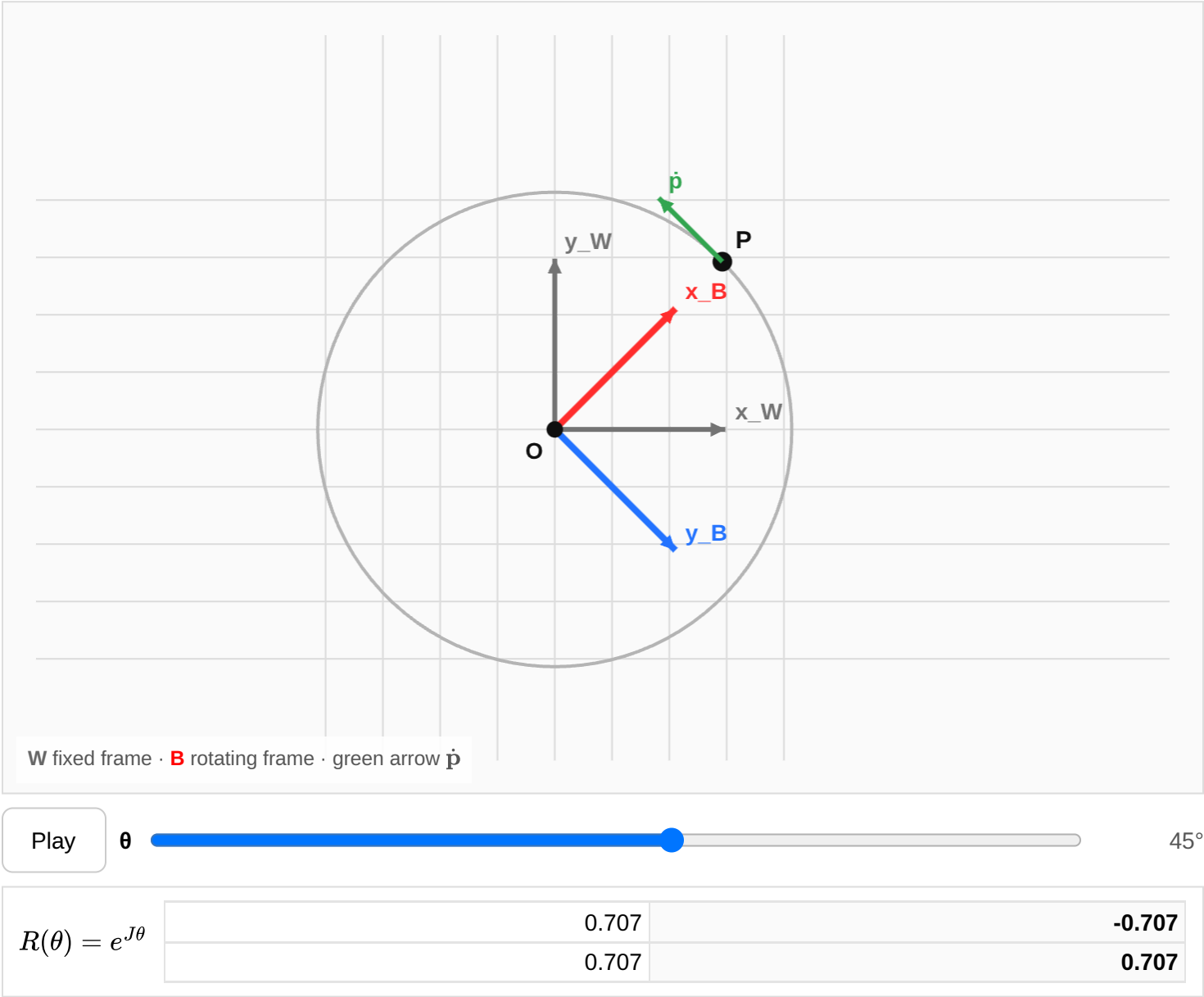
$$J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad J\mathbf{p} = \hat{z} \times \mathbf{p}.$$

Instantaneous motion: $\dot{\mathbf{p}}(t) = \omega J\mathbf{p}(t)$.

Accumulate small rotations:
 $\mathbf{p}(t + \Delta t) \approx (I + \omega J\Delta t)\mathbf{p}(t)$.

Take the continuous limit:

$$\mathbf{p}(t) = e^{\omega J t} \mathbf{p}(0) = e^{J\theta} \mathbf{p}(0), \quad \theta = \omega t.$$



Takeaway: the exponential is not arbitrary notation—it is what repeated infinitesimal rotations converge to.

The exponential answers a time-evolution question

If a matrix generator A produces the instantaneous change of a state, its exponential produces the finite change.

Begin with a constant linear differential equation.

$$\dot{X}(t) = AX(t), \quad X(0) = I$$

The generator A tells us the tangent direction at each instant.

The unique finite evolution is

$$X(t) = e^{At} = I + At + \frac{(At)^2}{2!} + \dots$$

For rigid motion, the generator lies in $\mathfrak{se}(3)$, while the result lies in $SE(3)$.

Takeaway: exponentiation converts an instantaneous motion rule into a finite motion.

Angular velocity describes how orientation changes

Its direction is the instantaneous rotation axis; its magnitude is the angular speed.

For a point $\mathbf{p}$ attached to a body rotating about an axis through the origin,

$$\dot{\mathbf{p}} = \boldsymbol{\omega} \times \mathbf{p} = [\boldsymbol{\omega}]_{\times} \mathbf{p}$$

Right-hand rule: curled fingers give positive rotation; the thumb gives $\boldsymbol{\omega}$.

The cross product becomes a linear operator

$$[\boldsymbol{\omega}]_{\times} = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix}$$

Because the matrix is skew-symmetric, $[\boldsymbol{\omega}]_{\times}^T = -[\boldsymbol{\omega}]_{\times}$.

A screw couples rotation and translation

A general instantaneous rigid motion combines rotation about an axis with translation along it; pure translation is the zero-angular-part case.

Let $\mathbf{r}$ point from the coordinate origin to a point on the screw axis, with unit axis direction $\boldsymbol{\omega}$ and pitch h .

$$\mathbf{v} = -\boldsymbol{\omega} \times \mathbf{r} + h\boldsymbol{\omega}$$

The radius from the axis back to the coordinate origin is $-\mathbf{r}$. The right-hand rule gives $\boldsymbol{\omega} \times (-\mathbf{r})$; $h\boldsymbol{\omega}$ adds translation along the axis.

Revolute

$$\|\boldsymbol{\omega}\| = 1, h = 0$$

Helical

$$\|\boldsymbol{\omega}\| = 1, h \neq 0$$

Prismatic

$$\boldsymbol{\omega} = \mathbf{0}, \|\mathbf{v}\| = 1$$

Here $\mathbf{v}$ is not generally the velocity of the body-frame origin; it is the linear part of the twist representation.

The hat operator turns six numbers into a generator

The matrix form is chosen so it acts on homogeneous points and exponentiates into a homogeneous transformation.

$$\xi = \begin{bmatrix} \omega \\ \mathbf{v} \end{bmatrix} \in \mathbb{R}^6$$

A six-vector is convenient for storage and Jacobians.

$$\hat{\xi} = \begin{bmatrix} [\omega]_{\times} & \mathbf{v} \\ \mathbf{0}^T & 0 \end{bmatrix} \in \mathfrak{se}(3)$$

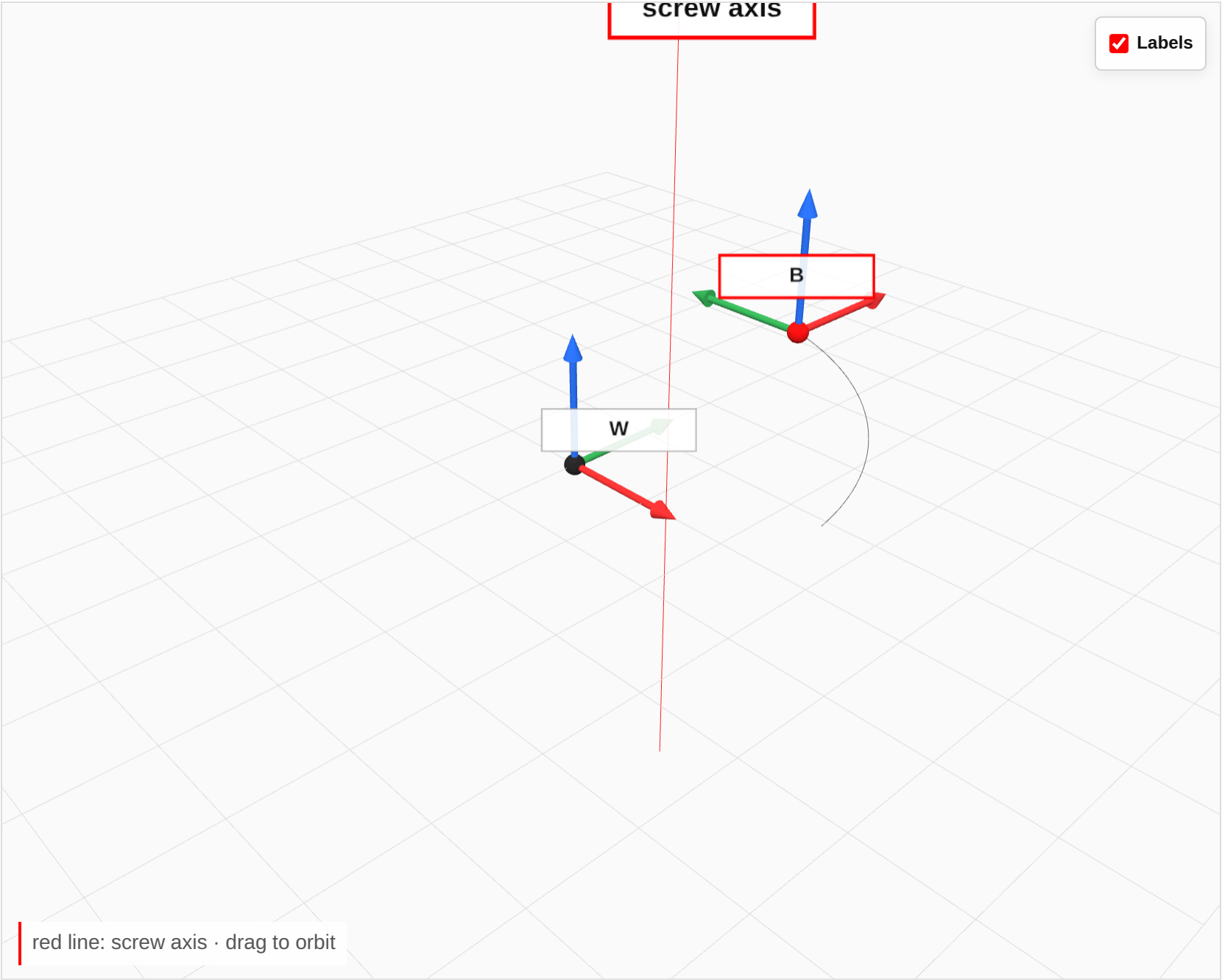
The hat matrix is convenient for applying the exponential map.

$$T(\theta) = e^{\hat{\xi}\theta} \in SE(3)$$

One twist generates a family of poses

Change angle and pitch. Watch rotation around the axis and translation along it remain coupled.

Explore: set pitch to zero, then increase it. Track which entries in the finite transformation change.



Screw parameters

θ 90°

pitch h 0.20

Twist

$\omega = [0, 0, 1]$ $v = [0.00, -0.65, 0.20]$

Finite motion $e^{\hat{\xi}\theta}$

0.000	-1.000	0.000	0.650
1.000	0.000	0.000	-0.650
0.000	0.000	1.000	0.314
0.000	0.000	0.000	1.000

The skew generator has only three independent powers

Let $K = [\boldsymbol{\omega}]_{\times}$, where $\|\boldsymbol{\omega}\| = 1$. We first simplify powers of K before expanding the exponential.

Apply K^2 to any vector $\mathbf{p}$.

$$K^2 \mathbf{p} = \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{p}) = \boldsymbol{\omega}(\boldsymbol{\omega}^T \mathbf{p}) - \mathbf{p}$$

The vector triple-product identity gives

$$K^2 = \boldsymbol{\omega} \boldsymbol{\omega}^T - I.$$

Multiply once more by K .

$$K^3 = K(\boldsymbol{\omega} \boldsymbol{\omega}^T - I) = (K\boldsymbol{\omega})\boldsymbol{\omega}^T - K.$$

Because $K\boldsymbol{\omega} = \boldsymbol{\omega} \times \boldsymbol{\omega} = \mathbf{0}$,

$$\boxed{K^3 = -K}, \quad \boxed{K^4 = -K^2}.$$

Consequence: every higher power reduces to either K or K^2 .

Regroup the exponential into odd and even powers

Substitute $K^3 = -K$ and $K^4 = -K^2$ into the Taylor series.

$$\begin{aligned}
 e^{K\theta} &= I + K\theta + \frac{K^2\theta^2}{2!} + \frac{K^3\theta^3}{3!} + \frac{K^4\theta^4}{4!} + \dots \\
 &= I + K \left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots \right) + K^2 \left(\frac{\theta^2}{2!} - \frac{\theta^4}{4!} + \frac{\theta^6}{6!} - \dots \right).
 \end{aligned}$$

Odd coefficient

$$\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots = \sin \theta$$

Even coefficient

$$\frac{\theta^2}{2!} - \frac{\theta^4}{4!} + \frac{\theta^6}{6!} - \dots = 1 - \cos \theta$$

Rodrigues' formula is the closed-form exponential

Insert the sine and cosine series, then interpret how the three terms act on a vector.

$$R(\theta) = e^{[\boldsymbol{\omega}]_{\times} \theta} = I + \sin \theta [\boldsymbol{\omega}]_{\times} + (1 - \cos \theta) [\boldsymbol{\omega}]_{\times}^2$$

$I\mathbf{p}$ starts from the original vector	$\sin \theta (\boldsymbol{\omega} \times \mathbf{p})$ moves in the tangential direction	$(1 - \cos \theta) \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{p})$ supplies the radial correction
--	--	--

Check: at $\theta = 0$, $R = I$; differentiating at zero gives $\left. \frac{dR}{d\theta} \right|_0 = [\boldsymbol{\omega}]_{\times}$.

Derive $G(\theta)$ from the exponential's translation block

Let $K = [\boldsymbol{\omega}]_{\times}$, with $\|\boldsymbol{\omega}\| = 1$, and $\hat{\boldsymbol{\xi}} = \begin{bmatrix} K & \mathbf{v} \\ \mathbf{0}^T & 0 \end{bmatrix}$. Expand the same matrix exponential used for rotation.

1 · Expand the block powers

$$\hat{\boldsymbol{\xi}}^2 = \begin{bmatrix} K^2 & K\mathbf{v} \\ \mathbf{0}^T & 0 \end{bmatrix}, \quad \hat{\boldsymbol{\xi}}^n = \begin{bmatrix} K^n & K^{n-1}\mathbf{v} \\ \mathbf{0}^T & 0 \end{bmatrix} \quad (n \geq 1).$$

$$e^{\hat{\boldsymbol{\xi}}\theta} = I_4 + \theta\hat{\boldsymbol{\xi}} + \frac{\theta^2}{2!}\hat{\boldsymbol{\xi}}^2 + \frac{\theta^3}{3!}\hat{\boldsymbol{\xi}}^3 + \dots$$

Read the upper-right block:

$$\mathbf{t}(\theta) = \underbrace{\left(\theta I + \frac{\theta^2}{2!}K + \frac{\theta^3}{3!}K^2 + \frac{\theta^4}{4!}K^3 + \dots \right)}_{G(\theta)} \mathbf{v}.$$

Each translation term has one fewer power of K . Equivalently, $G(\theta)$ is the term-by-term integral of $R(\theta)$.

2 · Reduce powers and recognize the series

Use $K^3 = -K$, $K^4 = -K^2$, $K^5 = K$, ...

$$G(\theta) = \theta I + \underbrace{\left(\frac{\theta^2}{2!} - \frac{\theta^4}{4!} + \frac{\theta^6}{6!} - \dots \right)}_{1 - \cos \theta} K + \underbrace{\left(\frac{\theta^3}{3!} - \frac{\theta^5}{5!} + \frac{\theta^7}{7!} - \dots \right)}_{\theta - \sin \theta} K^2.$$

$$\boxed{G(\theta) = \theta I + (1 - \cos \theta)K + (\theta - \sin \theta)K^2.}$$

The same result by integrating Rodrigues:

$$G(\theta) = \int_0^\theta e^{Ku} du = \int_0^\theta [I + \sin u K + (1 - \cos u)K^2] du.$$

$$e^{\hat{\boldsymbol{\xi}}\theta} = \begin{bmatrix} R(\theta) & G(\theta)\mathbf{v} \\ \mathbf{0}^T & 1 \end{bmatrix}.$$

Pure translation: if $\boldsymbol{\omega} = \mathbf{0}$, then $K = 0$, $G(\theta) = \theta I$, and $\mathbf{t} = \theta\mathbf{v}$.

Recover the velocity of the tool point

Let E be the moving tool frame and W the fixed world frame. Write $T = {}^W T_E$, $R = {}^W R_E$, and $\mathbf{p} = {}^W \mathbf{p}_E$, the position of the tool origin.

1 · Differentiate the pose

$$T = \begin{bmatrix} R & \mathbf{p} \\ \mathbf{0}^T & 1 \end{bmatrix}, \quad T^{-1} = \begin{bmatrix} R^T & -R^T \mathbf{p} \\ \mathbf{0}^T & 1 \end{bmatrix}.$$

$$\dot{T}T^{-1} = \begin{bmatrix} \dot{R}R^T & \dot{\mathbf{p}} - \dot{R}R^T \mathbf{p} \\ \mathbf{0}^T & 0 \end{bmatrix}.$$

Since $RR^T = I$, the matrix $\dot{R}R^T$ is skew-symmetric:

$$\dot{R}R^T = [{}^W \boldsymbol{\omega}_E]_{\times}.$$

2 · Read the spatial twist

$${}^W V_E^s = \begin{bmatrix} {}^W \boldsymbol{\omega}_E \\ {}^W \mathbf{v}_E \end{bmatrix}, \quad \widehat{{}^W V_E^s} = \dot{T}T^{-1}.$$

Its linear coordinate is therefore

$${}^W \mathbf{v}_E = {}^W \dot{\mathbf{p}}_E - {}^W \boldsymbol{\omega}_E \times {}^W \mathbf{p}_E.$$

V_E^s contains actual velocity rates. The earlier ξ_i describes a normalised joint screw; multiplying by its joint rate gives a velocity twist.

Velocity of the physical tool point

$${}^W \dot{\mathbf{p}}_E = {}^W \mathbf{v}_E + {}^W \boldsymbol{\omega}_E \times {}^W \mathbf{p}_E$$

Check: rotation about z_W through the world origin has ${}^W \mathbf{v}_E = \mathbf{0}$. At

${}^W \mathbf{p}_E = (r, 0, 0)^T$, with ${}^W \boldsymbol{\omega}_E = (0, 0, \Omega)^T$, the tool moves with

$${}^W \dot{\mathbf{p}}_E = (0, \Omega r, 0)^T.$$

One physical motion, different twist coordinates

The adjoint is the 6×6 coordinate-change matrix induced by a rigid transformation. It rotates both components and accounts for the shift of origin.

Notation

${}^A\xi_C = [{}^A\omega_C; {}^A\mathbf{v}_C]$: twist of body C , expressed in A and reduced at the coordinate origin O_A .

${}^AT_B = (R, \mathbf{t})$: pose of B expressed in A , with $R = {}^AR_B$ and $\mathbf{t} = {}^A\mathbf{t}_B$.

Ad_T : adjoint representation of T .

$$\text{Ad}_T = \begin{bmatrix} R & 0 \\ [\mathbf{t}]_{\times} R & R \end{bmatrix}, \quad {}^A\xi_C = \text{Ad}_{AT_B} {}^B\xi_C.$$

Why the lower-left block? Moving the coordinate origin changes linear velocity by $\mathbf{t} \times \omega$, even though the rigid-body motion is unchanged.

Change screw coordinates

Express a joint axis from its local URDF frame in world or tool coordinates.

Relate Jacobians

For ${}^sT_b(q)$, ${}^sJ_E = \text{Ad}_{{}^sT_b(q)} {}^bJ_E$. Both describe body E , with twists reduced at their respective coordinate origins.

Transform velocities

Move spatial or body twists between sensor, base, and end-effector frames.

Move forces dually

Wrenches use the inverse transpose: ${}^A\mathcal{F}_C = \text{Ad}_{AT_B}^{-T} {}^B\mathcal{F}_C$.

Express the twist at the tool point

The geometric tool twist stacks angular velocity and the actual velocity of the tool origin. We retain this lecture's angular-first order.

Reduce the twist at the tool point

Let $T = {}^W T_E$ and $\mathbf{p} = {}^W \mathbf{p}_E$.

$${}^W V_E^{\text{tool}} = \begin{bmatrix} {}^W \boldsymbol{\omega}_E \\ {}^W \dot{\mathbf{p}}_E \end{bmatrix} = \begin{bmatrix} I & 0 \\ -[\mathbf{p}]_{\times} & I \end{bmatrix} {}^W V_E^s.$$

The minus sign follows from $\boldsymbol{\omega} \times \mathbf{p} = -\mathbf{p} \times \boldsymbol{\omega}$.

Move to another attached point P

Let $\mathbf{r} = {}^W \mathbf{p}_P - {}^W \mathbf{p}_E$ point from E to P . Rigidity gives

$${}^W \dot{\mathbf{p}}_P = {}^W \dot{\mathbf{p}}_E + {}^W \boldsymbol{\omega}_E \times \mathbf{r}.$$

Angular velocity is shared by every point on the rigid tool.

Express the same tool-point velocity in E

With $R = {}^W R_E$, rotate both vectors into tool axes:

$${}^E V_E^b = \begin{bmatrix} R^T & 0 \\ 0 & R^T \end{bmatrix} {}^W V_E^{\text{tool}} = \begin{bmatrix} R^T {}^W \boldsymbol{\omega}_E \\ R^T {}^W \dot{\mathbf{p}}_E \end{bmatrix}.$$

This is the body twist. The same result follows directly from the pose derivative:

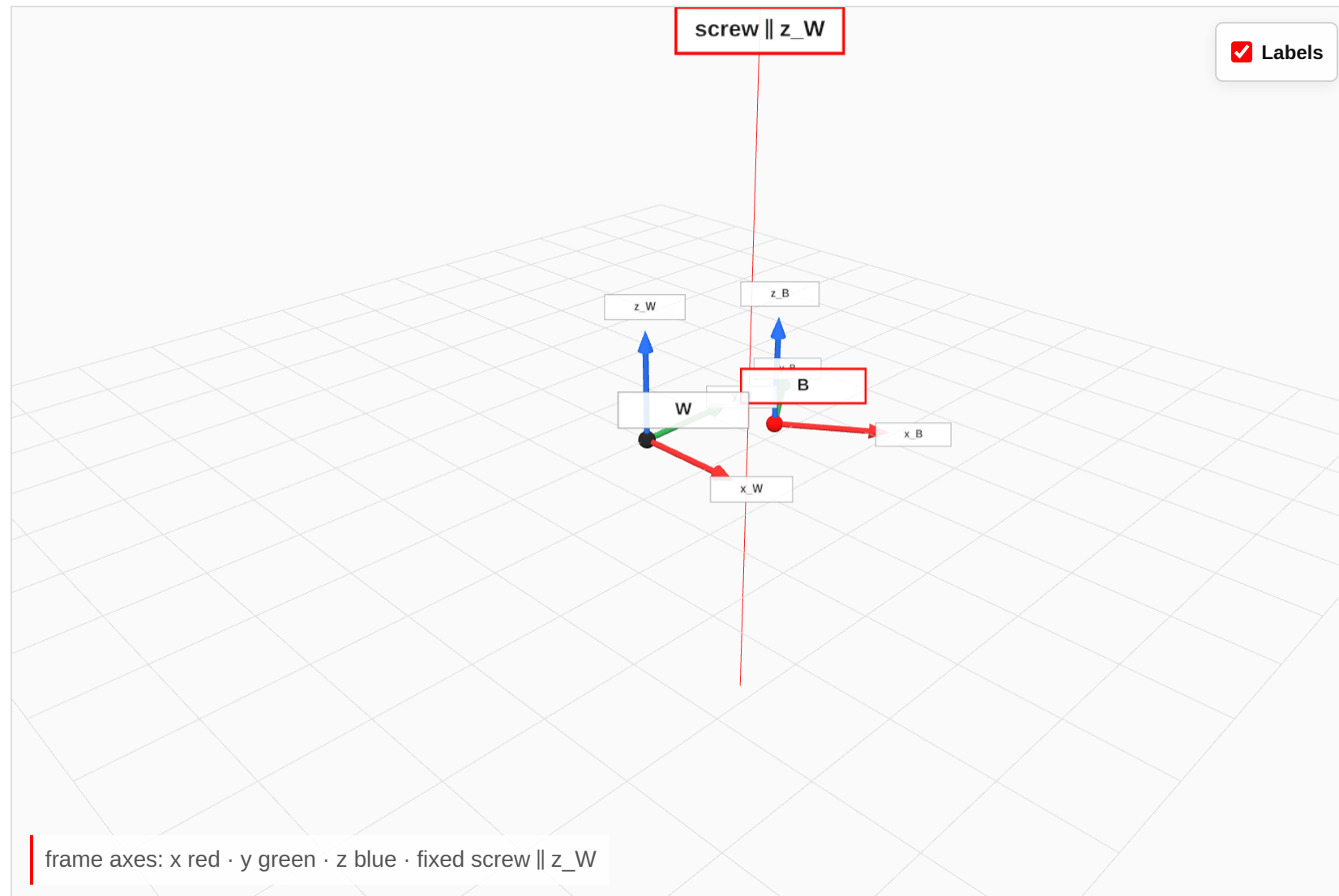
$$\widehat{{}^E V_E^b} = T^{-1} \dot{T}, \quad {}^E V_E^b = \text{Ad}_{T^{-1}} {}^W V_E^s.$$

$R^T {}^W \dot{\mathbf{p}}_E$ is the physical velocity expressed in moving axes. It is not the derivative of ${}^E \mathbf{p}_E = \mathbf{0}$.

Two distinct operations: changing the point adds a cross product to the linear velocity; changing axes at a fixed point rotates both velocity vectors. The adjoint combines these operations.

Move the observer, not the screw

The red screw i is parallel to z_W through ${}^W\mathbf{r}_i = (0.8, 0, 0)^T$. The sliders set ${}^W\mathbf{p}_B = (x, y, z)^T$ and the orientation of frame B ; the readouts express the same screw in each frame.



Pose ${}^W T_B$		
${}^W p_{B,x}$ [m]	<input type="range" value="0.40"/>	0.40
${}^W p_{B,y}$ [m]	<input type="range" value="0.70"/>	0.70
yaw [°]	<input type="range" value="35"/>	35°

Same screw, coordinates in W and B

Component order: $\xi = [\omega_x, \omega_y, \omega_z; v_x, v_y, v_z]$

$${}^W \xi_i = [0.00, 0.00, 1.00; 0.00, -0.80, 0.00]$$

$${}^B \xi_i = [0.00, 0.00, 1.00; -0.80, 0.07, 0.00]$$

Adjoint check

$${}^B \xi_i = \text{Ad}_{({}^W T_B)^{-1}} {}^W \xi_i$$

At zero yaw: ${}^B \mathbf{v}_i = [-y, x - 0.8, 0]^T$. Moving B along the world x -axis changes the local y -component; ${}^B \mathbf{v}_i = -{}^B \boldsymbol{\omega}_i \times {}^B \mathbf{r}_i$ couples the axes.

PoE stores one screw per joint and one home pose

We use the space form: every normalised screw is expressed in the fixed world frame at the home configuration and ordered from root to tool.

$${}^W\xi_i$$

Space screw of joint i ,
expressed in W , evaluated at
 $q = \mathbf{0}$.

$$q_i$$

Joint displacement: radians for
revolute, metres for prismatic.

$$M$$

End-effector pose WT_E at
 $q = \mathbf{0}$.

$${}^WT_E(q)$$

End-effector pose after applying
all joint motions.

$${}^WT_E(q) = e^{\widehat{{}^W\xi_1 q_1}} e^{\widehat{{}^W\xi_2 q_2}} \dots e^{\widehat{{}^W\xi_n q_n}} M$$

Order matters: the exponentials do not commute. A body-form PoE uses body screws and places M on the other side; do not mix the two forms.

Read each screw directly from the home geometry

For a revolute joint, let ${}^W\mathbf{r}_i$ point from the world origin to its axis and ${}^W\boldsymbol{\omega}_i$ give the unit axis direction.
Then ${}^W\mathbf{v}_i = -{}^W\boldsymbol{\omega}_i \times {}^W\mathbf{r}_i$.

Joint	${}^W\boldsymbol{\omega}_i$	Point ${}^W\mathbf{r}_i$ on axis	${}^W\mathbf{v}_i$
1	$[0, 0, 1]^T$	$[0, 0, 0.5]^T$	$[0, 0, 0]^T$
2	$[0, 1, 0]^T$	$[1, 0, 1]^T$	$[-1, 0, 1]^T$
3	$[0, 0, 1]^T$	$[3, 1.25, 1]^T$	$[1.25, -3, 0]^T$

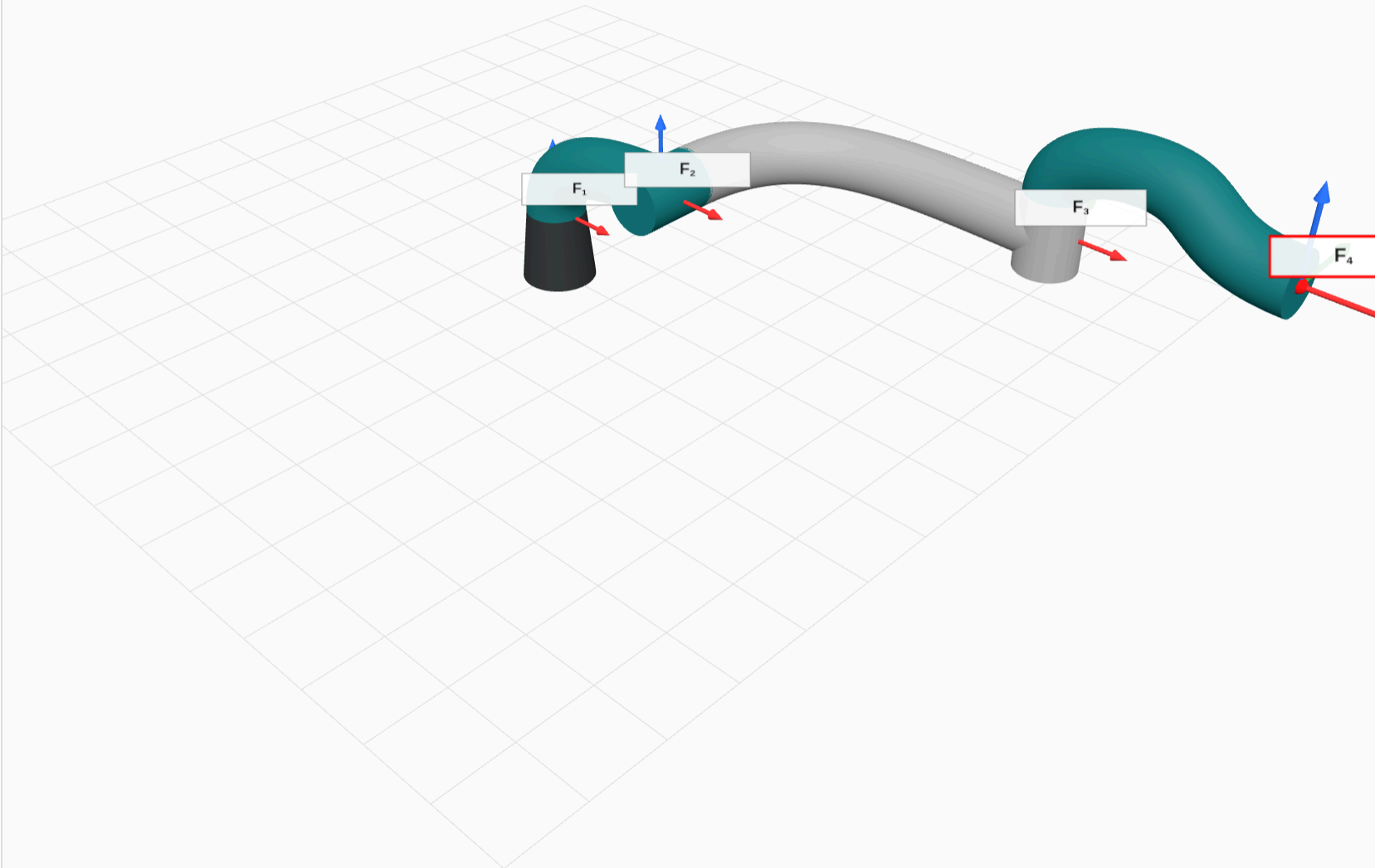
$$M = \begin{bmatrix} I & [4.5, 1.25, 1.25]^T \\ \mathbf{0}^T & 1 \end{bmatrix}$$

Compose the three joint exponentials

Move the joint coordinates and inspect each factor, the end-effector path, and the resulting ${}^W T_E(q)$.

☒ Labels

STL opacity 100%



custom_3R URDF/STL · space PoE · black curve: end-effector trace

Joint coordinates

q1

0.0°

q2

0.0°

q3

0.0°

☒ Keep home configuration as ghost

50%

Loaded custom_3R · 4 STL visuals

Ordered factors

$e^{\widehat{W\xi_1}q_1}$

$e^{\widehat{W\xi_2}q_2}$

$e^{\widehat{W\xi_3}q_3}$

M

Tool pose ${}^W T_E(q)$

1.000	0.000	0.000	4.500
0.000	1.000	0.000	1.250
0.000	0.000	1.000	1.250
0.000	0.000	0.000	1.000

Clear path

PoE computes motion; URDF communicates the mechanism

The equation and the description file preserve the same kinematic map while serving different consumers.

$${}^W T_E(q) = e^{\widehat{{}^W \xi_1} q_1} \dots e^{\widehat{{}^W \xi_n} q_n} M$$

PoE is compact and geometric, but it does not identify mesh files, collision shapes, masses, or software interfaces.

Minimal valid tree excerpt:

```
<robot name="custom_3r">
  <link name="base_link"/>
  <link name="tool0"/>
  <joint name="tool_fixed" type="fixed">
    <parent link="base_link"/>
    <child link="tool0"/>
  </joint>
</robot>
```

URDF is a machine-readable description format consumed by kinematics, visualisation, control, and simulation tools.

The PUMA 560 illustrates why URDF-to-DH conversion is geometric

URDF joint frames may be placed for CAD convenience; DH frames must satisfy geometric rules between consecutive axes.

URDF gives

- A parent-to-child joint origin $T_{\text{origin},i}$
- A joint axis expressed in the joint frame
- Visual frames independent of the kinematic edge

Standard DH requires

- z_{i-1} coincident with joint axis i
- x_i along a common normal
- Explicit signs, joint-zero offsets, base, and tool transforms

1 propagate axes to one frame

2 find common normals

3 choose signs and offsets

4 verify FK numerically

For the PUMA wrist, three axes intersect: the DH model is compact, but several valid frame choices can describe the same geometry.

Extract space screws by traversing the home geometry

No common-normal construction is required: express every movable joint axis in one reference frame and retain root-to-tool order.

1 · Traverse every edge At the declared joint zeros, propagate fixed and movable origins to obtain ${}^W T_{J_i}$.	2 · Transform and normalise ${}^W \mathbf{r}_i = {}^W \mathbf{t}_{J_i},$ ${}^W \mathbf{a}_i = {}^W R_{J_i} {}^{J_i} \mathbf{a}_i / \ {}^{J_i} \mathbf{a}_i \ .$	3 · Branch by joint type Revolute: $[{}^W \mathbf{a}_i; -{}^W \mathbf{a}_i \times {}^W \mathbf{r}_i].$ Prismatic: $[0; {}^W \mathbf{a}_i].$ Fixed: no screw.	4 · Store order and home pose Keep movable joints root-to-tool and set $M = {}^W T_E(0).$
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URDF origins + axes

$\implies$

$\{ {}^W \boldsymbol{\xi}_i \}_{i=1}^n, M$

Calibration offsets and mimic relations must be resolved consistently before the chosen $q = \mathbf{0}$ is called the PoE home configuration.

URDF describes trees well, but not every mechanism directly

URDF handles branched rooted trees well, but one child cannot have two parents—so closed kinematic loops cannot be represented directly.

Well represented

Serial robots, branched trees, mobile manipulators, sensors, rigid attachments, standard one-DOF joints.

Needs curated modelling

Parallel mechanisms and closed loops; rolling contact; antiparallelogram couplings; cable or tendon constraints; nonstandard multi-body joint laws.

Approximation strategies include mimic joints, equivalent trees plus external constraints, simulator-specific plugins, or a richer multibody description format.

Compact, geometric, and software-facing views are complementary

No single representation dominates every stage of robot design and validation.

DH

Compact kinematic structure

Four scalars per deliberately assigned frame step expose axis relationships and support symbolic analysis.

Under the chosen convention, $a_i = 0$ means consecutive axes intersect or coincide.

PoE

Geometric motion structure

Screw axes are coordinate-portable through the adjoint and connect naturally to velocities and the Jacobian.

$${}^W T_E(q) = e^{\widehat{{}^W \xi_1} q_1} \dots e^{\widehat{{}^W \xi_n} q_n} M$$

URDF

Software-facing description

Topology, frames, limits, geometry, and physical metadata can be consumed by modelling and simulation tools.

tree + geometry + physical metadata

A trustworthy forward model makes inverse kinematics the next question

01 URDF separates kinematics from appearance.

Joint origins and axes determine motion; visual origins place meshes independently.

02 Exponentials integrate normalised twists.

They turn one joint's local motion rule into a finite transformation.

03 PoE composes the root-to-tool motion.

With aligned zero, base, and tool conventions, it must agree with DH and URDF forward kinematics.

04 Inverse kinematics reverses the question.

Given a desired workspace pose T^* , Lecture 03 asks for every joint vector q satisfying ${}^W T_E(q) = T^*$ —and begins the study of multiplicity.

Check the geometry before dividing by a_1

The generic elimination step below requires a nonzero offset between the first two joint axes.

$$R = 2a_1E + F_3 \implies E = \frac{R - F_3}{2a_1}, \quad \boxed{a_1 \neq 0}$$

When $a_1 \neq 0$

The distance equation can be rearranged to isolate the signed radial component E . Continue the generic algebraic elimination with this expression.

For the iiwa 7, $a_1 = 0$

The first two axes intersect. Fixing original q_3 does not change that intersection, so the displayed division is undefined. Use the geometric distance and projection construction, then recover the angles with Cramer's rule.

This restriction concerns the displayed elimination formula; it does not rule out algebraic treatments of the intersecting-axis case. In Exercise 02, return to the tutorial tab and choose the route that respects the geometry.